

PATIENTTRIAGE.AI

Business Proposal

A practical, explainable decision-support layer for emergency-department triage

By:

Team: The 3 NON DOCS

Team members: Hridya Balachandran, K R Tanvi, Siju Niyati Samji

B. Tech Final-Year Students

From: IIT Gandhinagar

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1. Executive Summary

Emergency departments have to make high-stakes prioritisation decisions quickly, often before a complete patient history is available. The operational challenge is not simply deciding who is sickest; it is doing so consistently when information is incomplete, patients present differently across age groups, and demand can rise sharply.

PatientTriage.ai is designed as a decision-support assistant for the triage team. It gives each arriving or waiting patient an explainable severity recommendation, a visible confidence score, and the specific factors that drove the result. The clinician remains in control: recommendations can always be overridden, and every override is recorded with a reason.

What makes the approach different

The system is deliberately conservative under uncertainty. Missing information or an unclear presentation lowers confidence rather than being silently treated as 'no risk'. If confidence becomes low, the recommendation is moved one level more urgent. A critical vital sign independently forces at least an emergent Level 2 recommendation.

The current prototype demonstrates this approach with 18 simulated patients and a working Streamlit dashboard. It includes a live queue, patient-level explainability and override, surge simulation, an audit log, and design notes.

18	3	5	4
simulated patients	age bands	triage levels	working dashboard views

2. Problem Framing

2.1 The operational problem

Triage happens at the front door of the emergency department, where time pressure and incomplete information are normal. A patient may arrive without prior records, provide an incomplete description of symptoms, or look stable while an important warning sign is developing. At the same time, the department may move from a normal queue to a surge in a matter of hours.

This creates four practical problems:

- Inconsistent first-pass prioritisation when several patients present at the same time.
- Risk of under-triage when missing history or vague symptoms are interpreted as reassuring.
- A single adult-oriented threshold set can miss meaningful differences in children and older adults.
- Waiting patients can deteriorate after their initial assessment unless re-assessment is actively managed.

2.2 Design objective

Core objective

Help the triage team make a faster, more consistent first-pass assessment while keeping clinical judgement, explainability, and escalation under uncertainty at the centre of the workflow.

2.3 Why this is a decision-support problem, not an automation problem

The proposal intentionally does not replace the triage nurse. The system produces a recommendation that can be reviewed in seconds. The nurse can accept it, investigate further, or override it. This keeps accountability with the clinical team while giving them a structured second check.

3. Target Users & Jobs to Be Done

User	What they need	How PatientTriage.ai helps
Triage nurse	A quick recommendation with the reason visible immediately.	Shows severity, confidence and triggered flags in one patient view; clinical override remains available.

Charge nurse / surge coordinator	A view of the whole queue and patients who have waited too long.	Ranks the queue by severity and wait time and flags patients due for re-assessment.
ED medical director / compliance	Consistency, explainability and a defensible record of decisions.	Rules-based scoring and a timestamped override audit trail make decisions reviewable.
Hospital IT / CMIO	A deployable system that can start with partial data and scale.	Core engine works from intake data; integrations are phased instead of being a prerequisite for the pilot.

4. Solution Design

The prototype combines a transparent scoring engine with an operational dashboard. The design is intentionally simple: the system should be understandable by a clinician at a glance, not require them to interpret a black-box model.

4.1 End-to-end workflow

01	02	03	04	05
Intake	Score	Check uncertainty	Escalate	Monitor
Age, complaint, symptoms, vitals, history availability	Age-adjusted rules + red flags	Missing data and under-reporting signals reduce confidence	Critical vitals or low confidence move urgency upward	Queue position, wait window and re-assessment alerts

4.2 Explainable rules engine

The primary severity recommendation is generated by a weighted rules engine. This keeps the decision path visible and testable. A future ML model can be added as a secondary signal, but it is not allowed to become the sole or unexplained driver of severity.

Signal	How it is used	Why it matters
Age-adjusted vitals	Heart rate, respiratory rate, SpO ₂ , temperature and systolic BP are compared with pediatric, adult and geriatric bands.	The same reading can have different clinical meaning across age groups.
Red-flag symptoms	High-risk complaints such as chest pain, breathing difficulty, stroke symptoms, severe allergy or suicidal ideation add weight independently of vitals.	Captures 'looks sick' presentations before vital signs necessarily deteriorate.
Data completeness	Missing vitals reduce confidence; they are not treated as normal values.	Makes uncertainty visible instead of hiding it.
History availability	No prior record on file reduces confidence.	The system stays cautious when baseline information is unavailable.
Under-reporting language	Phrases such as 'vague pain', 'just feels off' or generalized weakness reduce confidence.	Supports escalation when symptom descriptions may be incomplete.

4.3 Age-adjusted thresholds

The engine separates patients into three bands: pediatric (<12), adult (12–64) and geriatric (65+). Each band has its own warning and critical ranges for the five tracked vital signs. This avoids applying one adult-centric threshold set to every patient.

4.4 Confidence is a visible output

Every recommendation includes a confidence percentage. Confidence starts high and is reduced by concrete uncertainty signals such as missing vitals, no history on file, under-reported symptom language, and geriatric presentations without pain reported. The prototype applies a floor of 25% so uncertainty is never hidden behind an artificially precise score.

4.5 Escalation-under-uncertainty logic

1. If any vital is outside its critical range, the patient is moved to at least Level 2.
2. If confidence falls below 60%, the patient is moved one severity level more urgent.

This is the central asymmetric-cost decision in the design. Over-triage consumes staff time; under-triage can have much more serious consequences. The system therefore accepts some additional escalation in exchange for a lower tolerance for uncertainty.

4.6 Waiting-room safety monitor

Severity is not a one-time decision. The dashboard continuously checks how long a patient has been waiting against a safe maximum for their current level. When the window is exceeded, the patient is flagged for re-assessment.

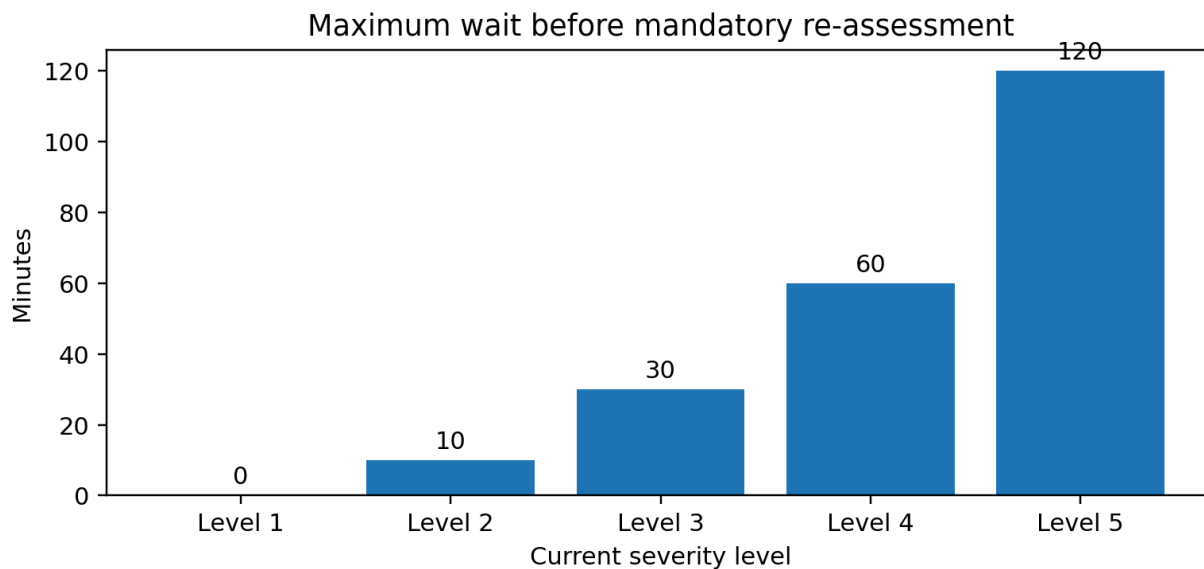


Figure 1. Re-assessment thresholds implemented in the prototype.

4.7 Surge behavior

The system does not lower its clinical thresholds when volume increases. Under a $3\times$ surge, the same scoring and escalation rules continue to apply. What changes is the operational load: more patients need monitoring and more re-assessment alerts can fire. The proposal therefore assigns surge-period alert ownership to the charge/surge nurse rather than allowing alerts to disappear into a crowded task list.

4.8 Clinician override and audit trail

Clinical judgement remains the final authority. An override requires a reason and records the patient ID, timestamp, original severity and confidence, new severity, clinician ID, and the system flags present at the time. This turns an override from an informal workaround into a reviewable clinical event.

5. Prototype Evidence

The current prototype contains 18 simulated patients designed to exercise the main edge cases: pediatric patients, geriatric patients, first-time/zero-history patients, ambiguous symptoms, and patients with clearly high-risk complaints.

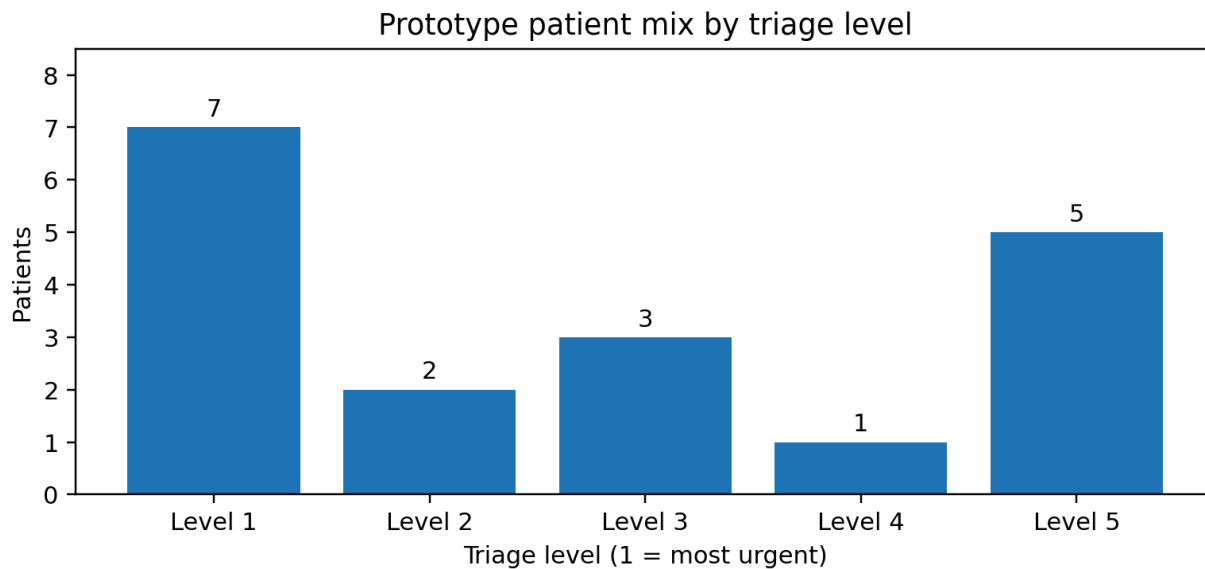


Figure 2. Severity distribution across the 18 simulated patients.

In the simulated panel, 7 patients are classified at Level 1, 2 at Level 2, 3 at Level 3, 1 at Level 4 and 5 at Level 5. These are prototype outputs from the supplied simulated dataset, not clinical performance claims.

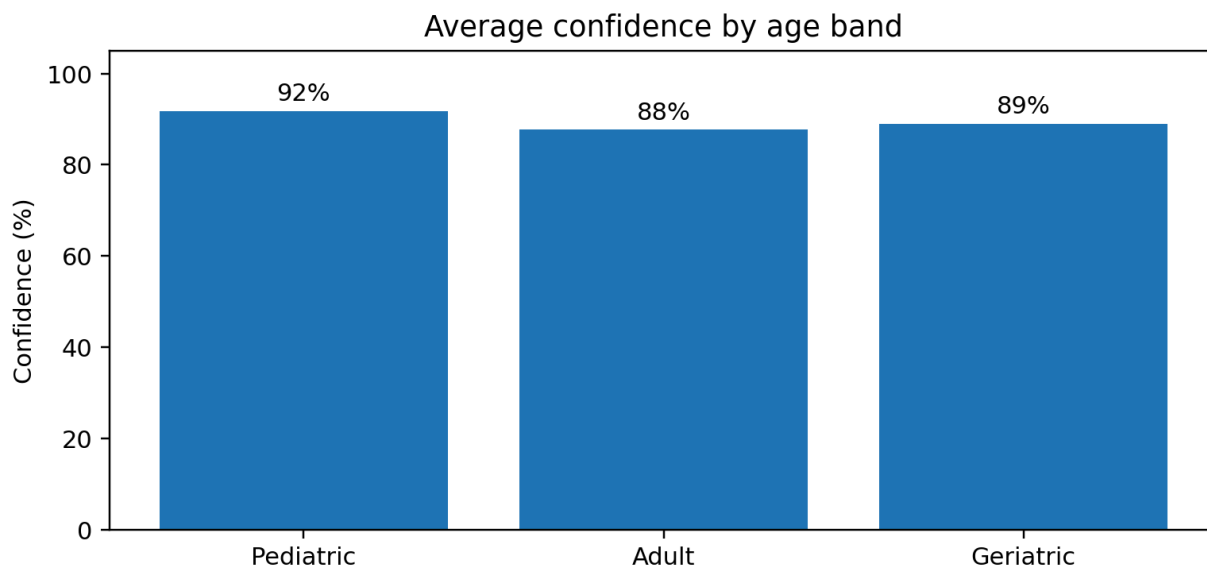


Figure 3. Average prototype confidence by age band.

The dataset also intentionally includes incomplete records: 6 of the 18 simulated patients have no prior history on file, and 2 patients have at least one missing vital. The purpose is to demonstrate that incomplete information changes confidence and can influence escalation rather than being silently interpreted as normal.

6. Business Case & Expected Impact

The business case is built around reducing avoidable risk and operational variability rather than promising an unrealistic 'AI replacement' for clinicians. The value comes from giving the team a consistent first-pass structure and making uncertainty visible.

Impact area	Mechanism	Priority
Reduce under-triage risk	Critical-vital and low-confidence escalation rules explicitly target the highest-cost failure mode.	High

Improve first-pass consistency	A common rules framework reduces variation in how the first assessment is structured.	High
Support faster queue management	Severity + wait time creates a single operational view for the charge/triage team.	Medium–High
Reduce review friction	Every override has a structured reason and timestamp for post-hoc review.	Medium–High
Deploy with partial data	The confidence layer is designed for patients with little or no prior record.	Medium
Create a foundation for future analytics	Structured scores, flags and overrides create a useful operational dataset for later improvement.	Medium

6.1 How value should be measured in the pilot

Metric	What to measure	Success signal
Time to first triage recommendation	Median time from complete intake data to a visible recommendation.	Lower without worsening safety indicators.
Agreement with clinician triage	Share of recommendations accepted or clinically aligned in shadow mode.	High agreement with explainable exceptions.
Under-triage events	Cases where the tool would have placed a patient less urgently than the final clinician decision.	Near-zero and investigated individually.
Over-triage rate	Share escalated by the system that clinicians subsequently judge as lower urgency.	Known, monitored trade-off rather than uncontrolled growth.
Re-assessment compliance	Share of flagged waiting patients re-assessed within the required window.	Consistently high compliance.
Override rate + reasons	How often clinicians override and why.	Stable rate with actionable reasons, not silent workarounds.
Alert burden	Alerts per shift and alerts requiring action.	Useful alerts without excessive fatigue.

7. Phased Roadmap

The rollout deliberately starts with observation and validation. Clinical authority is not handed to the system before the hospital has evidence that the recommendations are useful and safe in its own workflow.

Phase	Scope	Key deliverable / gate
Phase 0 — Prototype	Rules engine, simulated data, single-department dashboard, surge simulation and audit log.	Working prototype and documented safety logic.
Phase 1 — Pilot / shadow mode	Run alongside existing manual triage at one ED for 4–6 weeks. No clinical authority.	Compare recommendations with actual clinician decisions; tune thresholds and alert burden.
Phase 2 — Single-site live	Go live with EHR history lookup and full audit/compliance controls. Clinician override remains mandatory.	Formal safety review, security review and operational sign-off.
Phase 3 — Multi-site scaling	Configurable site thresholds, bed-management and staff-roster integration.	Repeatable deployment model for different ED sizes and workflows.
Phase 4 — Secondary ML signal	Add an ML risk model only as a secondary confidence input.	Calibration, drift and demographic-bias monitoring before any broader use.

8. Key Risks & Mitigations

Risk	Why it matters	Mitigation
Alert fatigue	Frequent re-assessment prompts can become background noise during a surge.	Route surge-period alerts to a dedicated charge-nurse role; tune wait thresholds during the pilot.
Staff distrust or workaround	A tool that slows nurses down will be bypassed.	Use shadow mode first, keep the explanation visible, and make overrides quick and legitimate.
Over-triage burden	Escalation bias can consume staff and bed capacity.	Track escalation-triggered cases and set an explicit over-triage ceiling during the pilot.
Bias in symptom weighting	Language such as 'vague pain' can be interpreted too aggressively.	Use these signals to reduce confidence, not to make assumptions about the patient's condition.
Hospital integration variance	Different hospitals have different EHRs, staffing and workflows.	Keep the core engine standalone and add integrations as separate modules.
PHI / compliance failure	Clinical data requires strict access and audit controls.	Encryption, role-based access, immutable audit logging and compliance review are go-live gates.
Model drift / future ML bias	A later ML signal may perform differently across sites or populations.	Introduce ML only as a secondary signal with calibration, drift and subgroup monitoring.

9. Operating Model & Data Assumptions

9.1 Human-in-the-loop operating model

- Triage nurse: enters/validates intake data, reviews the recommendation, and owns the clinical decision.
- Charge/surge nurse: monitors the queue and owns re-assessment alerts during high-volume periods.
- Medical director: owns clinical policy, threshold governance and periodic safety review.
- Hospital IT / CMIO: owns integration, access control, security and system availability.
- Compliance / quality team: reviews audit trails, overrides, safety events and pilot metrics.

9.2 Data and regulatory assumptions

- Prototype jurisdiction assumption: HIPAA / United States.
- All patient identifiers in the prototype are simulated.
- A production deployment should use hashed or tokenized identifiers in application logs and encrypt clinical data at rest and in transit.
- Role-based access should restrict PHI and audit-log visibility to authorised users.
- Secondary use of clinical data for model improvement should be handled under the hospital's applicable privacy and governance process.
- Audit records should follow the hospital's existing medical-record retention policy.

10. What the Prototype Already Demonstrates

Prototype view	What it demonstrates
Live Queue	Patients sorted by severity and wait time, with re-assessment alerts and simulated time controls.
Patient Detail & Override	Intake data, recommendation, confidence, explainability trace and clinician override.
Surge Simulation	3× volume simulation showing that clinical thresholds stay fixed while operational alert load increases.

Audit Log	Structured record of clinician overrides, including reason and system flags.
Design Notes	The safety, explainability, age-adjustment and uncertainty assumptions behind the system.

10.1 Suggested production dashboard layout

For production, the interface should stay close to the prototype's core idea: the triage nurse should not have to navigate through multiple screens to understand urgency. The queue should surface severity, confidence, wait time, and the most important reason for escalation. Detailed flags and audit information can remain one click deeper.

11. Recommendation

Recommended next step

Move to a 4–6 week shadow-mode pilot at one emergency department. Do not give the system clinical authority at first. Use the pilot to validate three things: whether recommendations are directionally useful, whether escalation creates an acceptable over-triage burden, and whether the re-assessment alerts improve queue discipline without causing alert fatigue.

If those three conditions are met, the next investment should be EHR integration and production-grade security/audit controls. The core scoring logic should remain transparent even as the product becomes more sophisticated.

Appendix A. Prototype Scoring Snapshot

Patient	Age	Band	Severity	Confidence	Score	Missing vitals
P001	34	Adult	Level 1	100%	115	0
P004	41	Adult	Level 1	69%	180	2
P007	6	Pediatric	Level 1	100%	115	0
P008	82	Geriatric	Level 1	95%	145	0
P010	1	Pediatric	Level 1	67%	80	1
P012	19	Adult	Level 1	85%	145	0
P017	88	Geriatric	Level 1	85%	80	0
P002	3	Pediatric	Level 2	100%	60	0
P003	78	Geriatric	Level 2	85%	60	0
P006	55	Adult	Level 3	75%	30	0
P014	29	Adult	Level 3	75%	40	0
P018	22	Adult	Level 3	85%	35	0
P013	71	Geriatric	Level 4	95%	20	0
P005	27	Adult	Level 5	100%	0	0
P009	45	Adult	Level 5	100%	0	0
P011	68	Geriatric	Level 5	85%	0	0
P015	8	Pediatric	Level 5	100%	0	0
P016	52	Adult	Level 5	100%	0	0

Note: This appendix reflects the supplied simulated dataset and the current prototype rules. It is included for transparency, not as evidence of clinical accuracy or real-world outcomes.

Appendix B. One-page proposal logic

Question	Answer
Problem	ED triage must prioritise quickly with incomplete information and changing queue conditions.

Solution	Explainable, age-adjusted, uncertainty-aware triage decision support with a live queue and re-assessment monitor.
Users	Triage nurses, charge/surge nurses, ED leadership/compliance, hospital IT/CMIO.
Business value	Lower under-triage risk, more consistent first-pass triage, better queue visibility and stronger auditability.
Rollout	Prototype → shadow pilot → single-site live → multi-site → optional secondary ML signal.
Key guardrail	The clinician always retains final authority; the system escalates rather than silently downgrading under uncertainty.